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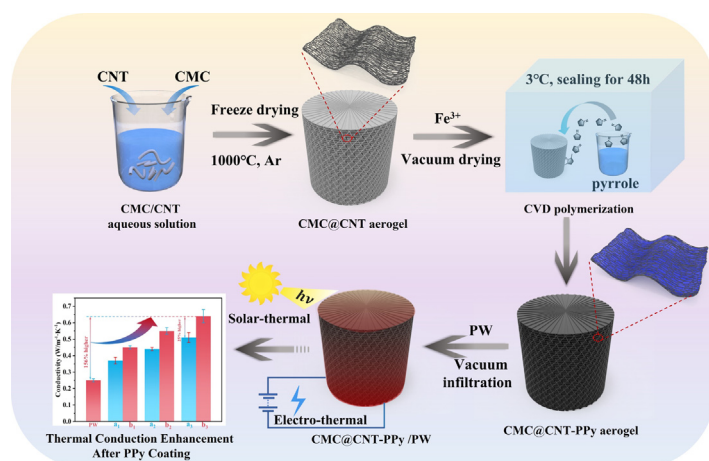
Polypyrrole coated carbon nanotube aerogel composite phase change materials with enhanced thermal conductivity, high solar-/electro-thermal energy conversion and storage

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GRAPHICAL ABSTRACT



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ABSTRACT

Phase change materials (PCMs) have been widely investigated as promising thermal management materials due to their high thermal storage capacity, satisfactory heat transfer rate and multi-responsive energy conversion and storage characteristics. In this work, a shape-stabilized solar-/electro- responsive thermal energy capture and storage system is proposed involving polypyrrole (PPy)-deposited carbon nanotubes (CNT) heterogeneous porous aerogel as a supporting matrix and the paraffin wax (PW) as a PCM. The composite PCMs obtained via integration of PW into aerogel supports present a relatively high thermal storage density of 160.9 J/g and outstanding phase transition stability even after 100 heating-cooling cycles. Furthermore, great enhancement of thermal conductivity ($0.64 \text{ W/m}^{-1}\cdot\text{K}^{-1}$, 2.56 times that of PW) is achieved in the composite PCMs by inducing PPy coating as a binder in the gap between CNTs. The mechanism of heat transport enhancement is explored by molecular dynamics simulation. It

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